Prodigy Anesthesia Quick Review

Benzodiazepines

Where in the brain do benzodiazepines exert their action?	The sedative effects are exerted on receptors in the cortex of the brain, amnesia is mediated by receptors in the forebrain and hippocampus, and the anxiolytic effects are exerted in the amygdala, hippocampus, and limbic system. *Ouellette RG, Joyce JA.Pharmacology for Nurse Anesthesiology. Sudbury, MA: Jones and Bartlett Learning; 2011: 53.*
How do benzodiazepines exert their action on GABA receptors?	Specific benzodiazepine receptor sites exist on a protein complex that contains binding sites for benzodiazepines, GABA, ethanol, barbiturates, and a chloride channel. When these receptor sites are occupied by one of the drugs listed above, the GABA receptor increases the frequency of chloride channel opening which results in hyperpolarization of the postsynaptic membrane and inhibition of neuronal transmission. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 480-481.*
How do benzodiazepines affect CMRO2 and cerebral blood flow?	They decrease CMRO2 and cerebral blood flow in a manner similar to propofol and thiopental. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
How are benzodiazepines metabolized?	Benzodiazepines are metabolized via hepatic oxidation and glucuronide conjugation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
How is the metabolism of benzodiazepines affected by hepatic disease?	They undergo oxidation, which is impaired in the presence of hepatic disease. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
How do benzodiazepines affect upper airway reflexes?	Benzodiazepines decrease the swallowing reflex and the upper airway reflexes. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
How do diazepam, lorazepam, and midazolam compare with respect to duration of action?	Diazepam is long-lasting with an elimination half-life of 36-50 hours, lorazepam is of intermediate duration with an elimination half-life of 10-22 hours, and midazolam is short-acting with an elimination half-life of about 2 hours. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 117.*

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Prodigy Anesthesia Quick Review

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How do the formulations of midazolam, lorazepam, and diazepam differ? How does this affect pain on injection?	Lorazepam and diazepam are not water soluble. They are formulated in propylene glycol which can cause pain on injection. Midazolam is water-soluble and formulated in an aqueous solution that causes minimal pain on injection. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 487-488.*
What is the typical IV or IM premedication dose of midazolam? What is the typical PO premedication dose of midazolam?	The dose for IV or IM midazolam is 0.02-0.04 mg/kg compared to 0.4 to 0.8 mg/kg for the PO dose. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
How does midazolam affect cerebral autoregulation?	Midazolam preserves the ability of the blood vessels to vasoconstrict in response to a decrease in blood flow. Despite a decrease in cerebral perfusion pressure, the intracranial pressure would be relatively unaffected. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
Can midazolam be used to prevent seizures?	Yes. Midazolam, lorazepam, and diazepam all increase the seizure threshold. *Miller RD. Miller's Anesthesia. 8th ed. Philadelphia: Elsevier Saunders; 2015: 840.*
How is the duration of action of diazepam affected by aging? Renal disease? Hepatic disease?	The duration of action of diazepam is prolonged in the elderly and in patients with hepatic or renal disease and doses should be decreased accordingly. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 699.*
How does diazepam affect the carbon dioxide response curve?	Diazepam decreases the slope of the CO2 response curve. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 119.*
How does diazepam affect minute ventilation?	Diazepam decreases minute ventilation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 119.*
What is the only benzodiazepine antagonist available in the United States?	Flumazenil *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 119.*

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Prodigy Anesthesia Quick Review

Benzodiazepines

How does flumazenil reverse the effects of benzodiazepines?	It is structurally similar to the benzodiazepines, but a phenyl group is replaced with a carbonyl group which makes the drug essentially inert, but still able to bind with the benzodiazepine receptor. It acts as a competitive antagonist. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488-489.*
How is flumazenil metabolized?	Flumazenil is rapidly metabolized in the liver and excreted in the urine. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 489.*
What is the elimination half-life of flumazenil and what is the significance of this?	It has a half-life of about 1 hour, so it is possible that re-sedation can occur with benzodiazepines that half a longer half-life than this. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 119.*
What is the typical dose of flumazenil and how long would you expect its effects to last?	Flumazenil 1-3 mg will reverse the effects of benzodiazepines for about 45-90 minutes. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*
What are the cardiovascular effects of flumazenil?	Flumazenil is not associated with any significant cardiovascular changes or stress response. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 119.*
How does flumazenil affect intracranial hemodynamics?	Flumazenil does not affect CMRO2 or cerebral blood flow. There is evidence that it can increase intracranial pressure when administered in head-injured patients. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 488.*

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